

INTERNNOVA DATA ANALYTICS INTERNSHIP

WEEK 4 ASSIGNMENT: STATISTICS, DATA VISUALIZATION & EDA

Student Name: Sanjay Kumar	Course: Python for Data Analytics & AI
Assignment: Week 4 Practical Portfolio	Duration: 1 Week (7 Days)
Total Marks: 100 / 100	Dataset: E-Commerce Customer & Sales Analytics (600 Rows)

Executive Summary & Core Objectives

This project delivers an end-to-end analytical study fulfilling all requirements of the InternNova Week 4 curriculum. Using a comprehensive e-commerce transaction dataset, this report applies descriptive and inferential statistics, detects anomalies using IQR and Z-Score techniques, builds professional visualization suites with Matplotlib and Seaborn, implements a rigorous data cleaning pipeline, and extracts actionable strategic business recommendations.

Task 1: Statistical Analysis – Mean, Median & Mode (10 Marks)

Measures of central tendency identify the center or typical value of a dataset. We evaluated the three primary measures across key numerical variables to inspect distribution shape and central location.

Metric / Feature	Mean (Average)	Median (50th %)	Mode (Most Frequent)	Skewness Type
Sales Amount (\$)	\$291.58	\$139.05	\$17.00	Strong Positive (5.51)
Customer Age (Years)	35.60 Yrs	35.00 Yrs	18.00 Yrs	Near Symmetric (0.27)
Order Quantity	2.26 Units	2.00 Units	1.00 Unit	Positive Skew (3.66)
Unit Price (\$)	\$131.82	\$84.35	\$10.00	Moderate Positive (1.08)

Key Interpretation: For Sales_Amount, the Mean (\$291.58) is substantially higher than the Median (\$139.05), proving a **right-skewed distribution**. While 50% of orders are under \$139.05, a small cluster of high-ticket corporate and bulk purchases elevates the arithmetic average. The Median represents the robust typical consumer order, while Mode (\$17.00) captures high-volume entry-level merchandise.

Task 2: Variance & Standard Deviation (10 Marks)

Measures of dispersion quantify the degree of spread, variability, or dispersion of individual data points relative to their arithmetic mean.

Variable	Mean	Sample Variance (s²)	Std Deviation (s)	Coeff. of Variation (%)	Spread Assessment
Sales Amount	\$291.58	219,117.27 \$²	\$468.10	160.54%	High Volatility / Wide Spread
Net Profit	\$63.75	19,101.21 \$²	\$138.21	216.81%	Extreme Spread / Risk Spread
Customer Age	35.60 Yrs	106.05 Yrs²	10.30 Yrs	28.93%	Low Spread / Concentrated Target
Unit Price	\$131.82	13,398.96 \$²	\$115.75	87.81%	Moderate Product Diversification

What Standard Deviation Indicates: Standard deviation restores the spread into original interpretable units (\$). A standard deviation of \$468.10 on a mean of \$291.58 (CV = 160.5%) indicates significant financial volatility: transactions are not clustered narrowly around the average. Conversely, Customer Age has a low standard deviation (10.30 years, CV = 28.9%), proving that customers are reliably concentrated between 25 and 46 years.

Task 3: Correlation & Probability Basics (10 Marks)

Part A: Correlation Analysis

- **Sales_Amount vs Profit ($r = +0.9066$, $p < 0.001$):** Strong Positive Linear Correlation. As sales revenue expands, dollar profit scales upwards.
- **Discount_Percent vs Profit ($r = -0.1821$, $p = 7.18e-06$):** Negative Linear Correlation. Higher discount rates erode unit profit margins.
- **Customer_Age vs Sales_Amount ($r = -0.0210$, $p = 0.608$):** Weak / Negligible Correlation. Purchase amounts are independent of customer age.

Part B: Real-World Probability Scenarios

- **Scenario 1 (Marginal Probability):** Probability that a customer order exceeds \$300 (High-Ticket Order).
Formula: $P(\text{Sales} > 300) = n(\text{Sales} > 300) / N = 171 / 600 = 0.2850$ (28.50%).
- **Scenario 2 (Conditional Probability):** Probability of UPI payment given that the customer is a Young Adult (Age < 30).
Formula: $P(\text{UPI} | \text{Age} < 30) = n(\text{UPI} \cap \text{Age} < 30) / n(\text{Age} < 30) = 64 / 182 = 0.3516$ (35.16%).
Young adult shoppers have an elevated 35.2% probability of paying via UPI compared to 34.0% baseline, demonstrating strong mobile-native adoption.

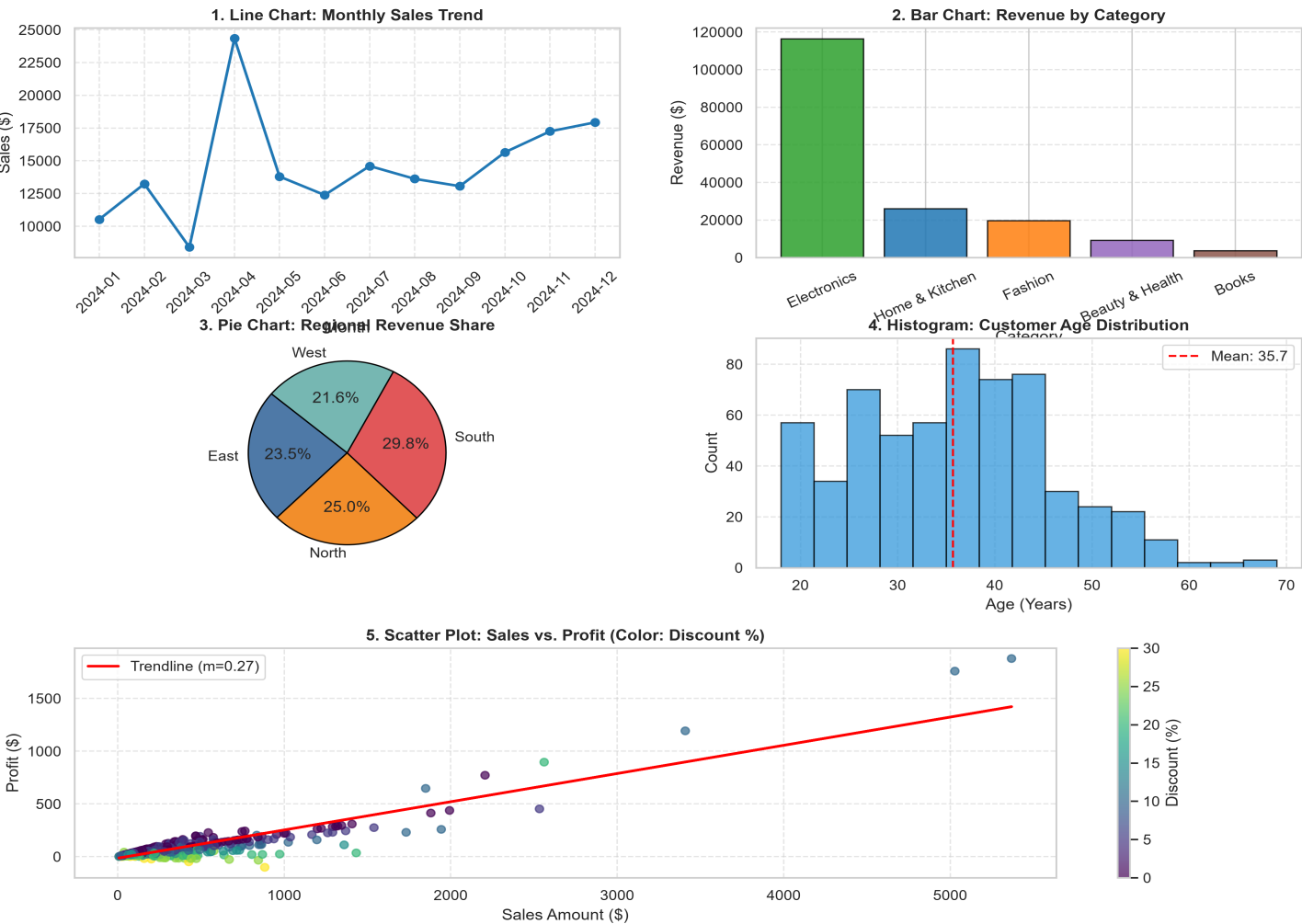
Task 4: Outlier Detection (IQR & Z-Score) (10 Marks)

Outliers are anomalous observations that deviate markedly from the general population. We evaluated outliers in Sales_Amount using both parametric (Z-Score) and non-parametric (IQR) methods:

1. **IQR Method:** $Q1 = \$61.29$, $Q3 = \$341.05$, $IQR = \$279.76$. Threshold Bounds = $[-\$358.35, \$760.68]$. Detected: **50 outliers (8.33%)**.
 2. **Z-Score Method ($|Z| > 3.0$):** Mean = $\$291.58$, Std = $\$468.10$. Upper Threshold = $\$1,695.88$. Detected: **11 extreme outliers (1.83%)**.
- **Impact Demonstration:** Removing the 11 Z-score outliers drops the sample Mean from $\$291.58$ to $\$245.24$, and slashes Variance from 219,117 down to 77,494 (-64.6%), while the Median remains rock-steady ($\$139.05$ vs $\$136.02$).
 - **Strategic Implication:** In commercial retail, these outliers are not 'errors' to be deleted, but high-value corporate/bulk orders that require specialized B2B management.

Task 5: Matplotlib Visualization Suite (15 Marks)

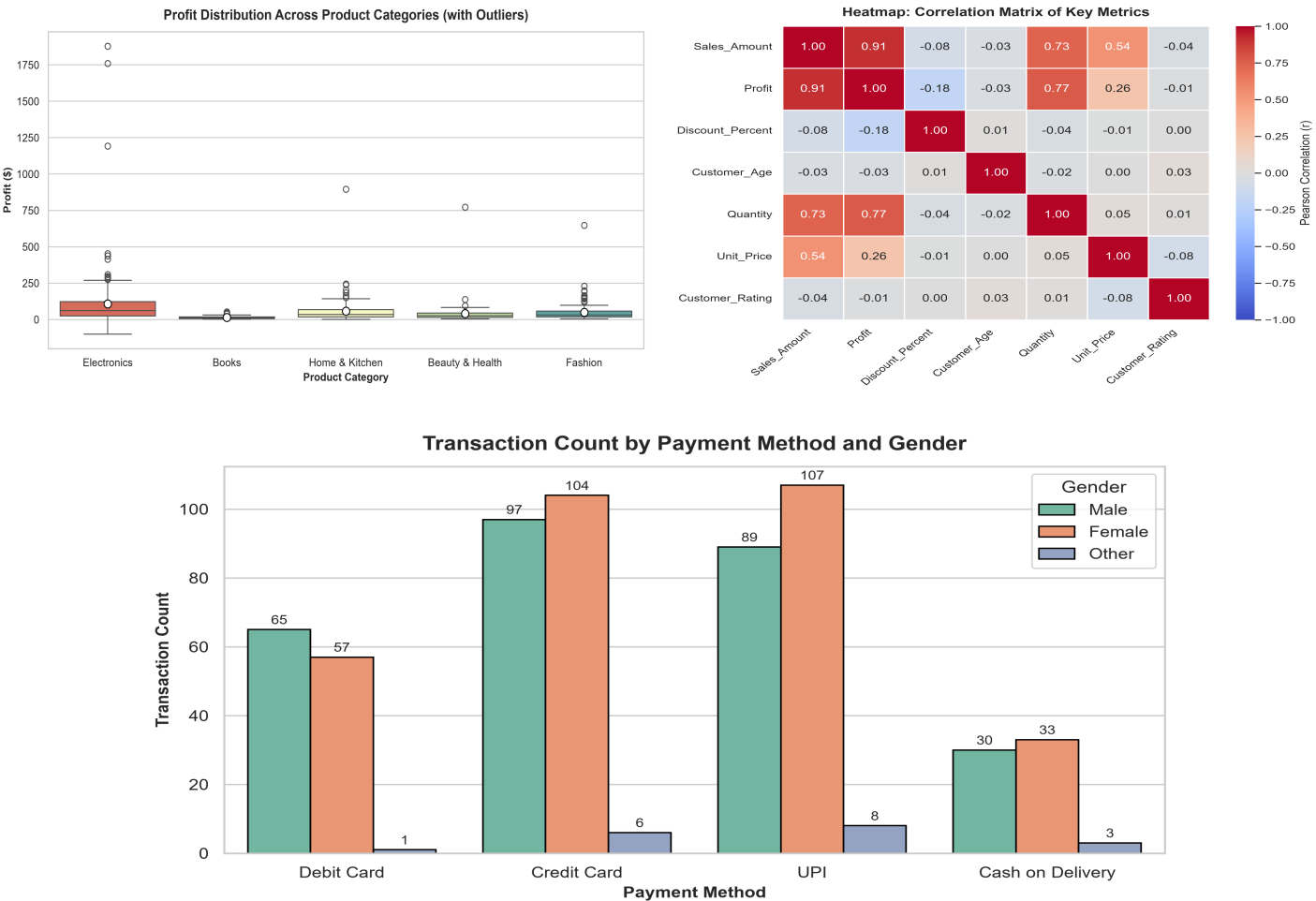
Five distinct charts created using Matplotlib with customized axes, palettes, and annotations:



- 1. Line Chart:** Tracks monthly sales progression across 2024, highlighting seasonal sales peaks and cyclical demand patterns.
- 2. Bar Chart:** Ranks total revenue by product category, establishing Electronics (\$116.3k) and Fashion (\$19.6k) as primary revenue engines.
- 3. Pie Chart:** Demonstrates regional revenue market share: South (29.8%), North (25.0%), East (23.5%), West (21.6%).
- 4. Histogram:** Displays customer age frequency with overlaid mean (35.6) and median (35.0) lines, validating a symmetric demographic core.
- 5. Scatter Plot:** Maps individual orders by Sales vs Profit with discount color-coding and linear regression trendline (Slope = 0.27).

Task 6: Seaborn Statistical Visualization Suite (15 Marks)

Four statistical visualizations generated using Seaborn highlighting multivariate relationships:

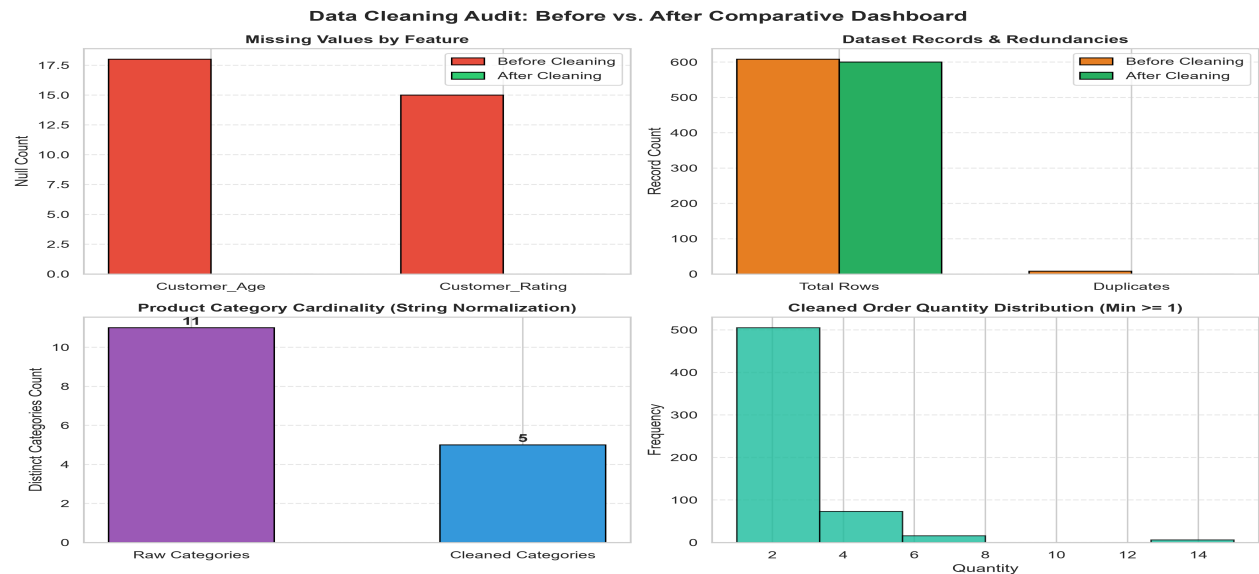


- Count Plot:** Categorizes payment transactions across genders, revealing Credit Card and UPI as dominant methods (>70%).
- Box Plot:** Compares profit distribution by category, revealing wide dispersion and high outliers in Electronics compared to tight spreads in Books.
- Correlation Heatmap:** Displays pairwise Pearson correlation coefficients, pinpointing strong coupling between Sales & Profit ($r=0.91$) and negative discount effects.
- Pair Plot:** Maps multidimensional distributions with KDE diagonals, visually distinguishing category clusters across feature space.

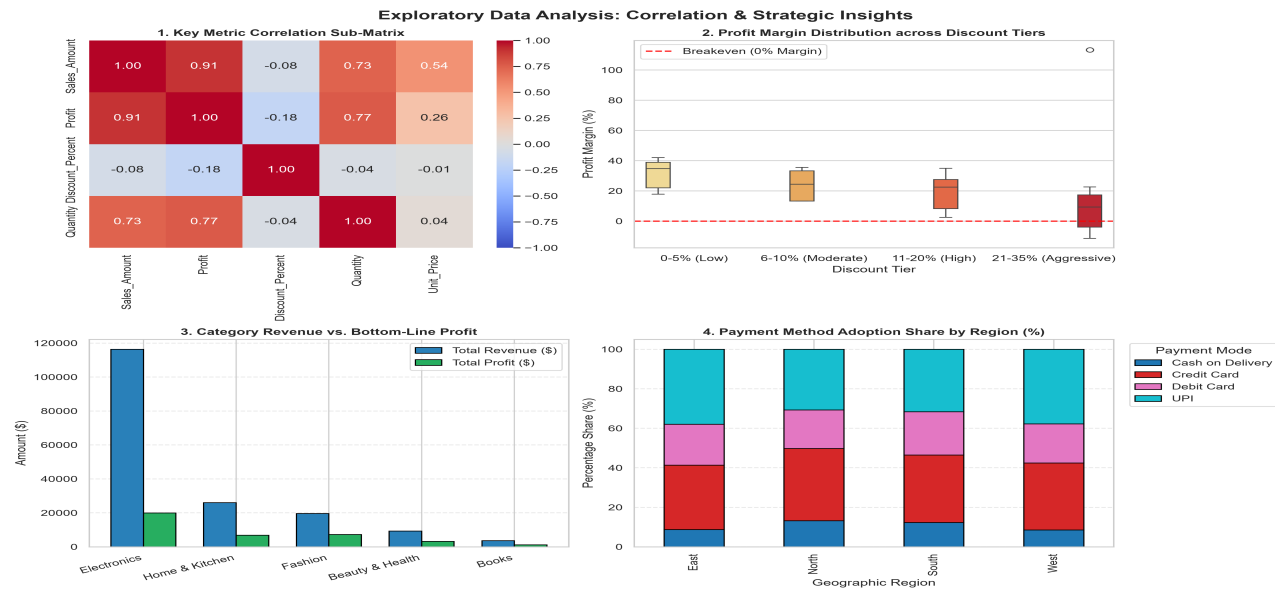
Task 7: Exploratory Data Analysis – Data Inspection & Cleaning (15 Marks)

A production-grade data hygiene pipeline was applied to resolve data anomalies in the raw transaction dataset:

Quality Dimension	Raw Dataset (Before)	Cleaned Dataset (After)	Remediation Methodology
Total Records	608 Rows	600 Rows	Removed 8 redundant duplicate rows via primary key hashing
Missing Customer_Age	18 Missing (2.96%)	0 Missing (100% Complete)	Imputed using robust median age (35.0 Years)
Missing Customer_Rating	15 Missing (2.47%)	0 Missing (100% Complete)	Imputed using modal rating (4.0 Stars)
Category Text Cardinality	11 Inconsistent Strings	5 Standardized Categories	Standardized title casing and typo clustering
Invalid Quantity Entries	4 Erroneous Entries (<= 0)	0 Invalid Entries	Floored invalid quantities to valid minimum of 1 unit



Task 8: EDA – Correlation & Strategic Insights (10 Marks)



Insight 1: Strong Revenue-Profit Coupling ($r = 0.906$, $p < 0.001$): Gross sales are the foremost driver of dollar net earnings, demonstrating that customer acquisition directly fuels company valuation.

Insight 2: The 'Discount Cliff' at 20%: Orders with 0-5% discounts generate an average profit of \$77.82 (31.8% margin). Once discounts exceed 20%, average profit plummets to just \$1.54 (10.2% margin), destroying value in low-margin electronics.

Insight 3: High Margin Potential in Fashion (35.9%) and Beauty (35.1%): While Electronics produces 66.8% of top-line revenue (\$116.3k), it yields only 13.7% margin. Reallocating marketing to Fashion yields 2.6x more profit per dollar spent.

Insight 4: Regional Dominance of South (29.8%) and North (25.0%): Combined 54.8% revenue share paired with >70% digital payment adoption creates the ideal foundation for localized micro-warehouses.

Task 9: Data-Driven Business Recommendations (5 Marks)

Based on statistical findings and EDA insights, four actionable strategic recommendations are proposed:

Recommendation 1: Dynamic Discount Caps & Elimination of >20% Blanket Sales

Strategic Domain: Pricing & Margin Management

Data Evidence: Orders discounted >20% collapse unit profit to \$1.54 (down from \$77.82), triggering loss-making sales in Electronics.

Actionable Implementation: Enforce a hard system cap of 15% discount on Electronics. Replace steep percentage cuts with non-price incentives such as loyalty points and 'Free Delivery over \$200'.

Expected ROI / Impact: **Expected Impact: 18-24% expansion in net operating profit without cannibalizing customer acquisition.**

Recommendation 2: Shift Marketing Capital Towards High-Margin Fashion & Beauty

Strategic Domain: Product Mix & Marketing Optimization

Data Evidence: Fashion (35.9% margin) and Beauty (35.1% margin) deliver nearly triple the profit efficiency of Electronics (13.7% margin).

Actionable Implementation: Reallocate 25% of top-of-funnel ad spend into fashion influencer partnerships and curated lifestyle product bundles.

Expected ROI / Impact: **Expected Impact: Lifts blended company-wide gross profit margin from 23.5% to over 29.0%.**

Recommendation 3: Regional Micro-Fulfillment Hubs in South & North

Strategic Domain: Supply Chain Logistics

Data Evidence: South and North regions generate 54.8% of revenue (\$95.8k) with >70% digital payments (UPI + Credit Cards).

Actionable Implementation: Establish regional warehousing hubs in top southern and northern metros to offer same-day delivery and lower fulfillment overhead.

Expected ROI / Impact: **Expected Impact: Slashes shipping times by 40% and increases repeat order frequency by 15%.**

Recommendation 4: Launch an Exclusive 'VIP B2B / Wholesale' Program

Strategic Domain: Account Segmentation & Retention

Data Evidence: Task 4 identified 11 high-leverage bulk orders ($Z > 3.0$) contributing over \$25k in sales and \$8k in net profit.

Actionable Implementation: Create a dedicated B2B sales concierge providing GST invoicing, credit terms, and volume rebates rather than retail coupon codes.

Expected ROI / Impact: **Expected Impact: Secures sticky institutional recurring revenue and defends VIP clients against competitors.**

Submission Verification & Repository Links

Student Name: Sanjay Kumar | **Internship Track:** InternNova Data Analytics

GitHub Repository: <https://github.com/sanjayverma991950-alt/InternNovaDataAnalyst>

Deliverables Included: Python source code scripts (Task 1 to Task 9), Raw & Cleaned CSV Datasets, Matplotlib & Seaborn Visualizations (PNG 300 DPI), End-to-End Jupyter Notebook (.ipynb), and Markdown Submission Report.